Historical Analysis of the Egyptian Exchange EGX30 Index (2014–2026): Exploratory Data Analysis and ARIMA Forecasting

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Abstract—*This paper presents a comprehensive historical analysis of the Egyptian Exchange EGX30 index over a 12-year period spanning January 2014 to May 2026, encompassing 3,000 trading days. The study employs exploratory data analysis (EDA) to examine return distributions, volatility clustering, drawdown behavior, volume dynamics, autocorrelation structure, and calendar anomalies including the Ramadan effect and day-of-week patterns. Technical indicators including SMA, EMA, RSI, MACD, Bollinger Bands, and ATR are computed as part of a 33-feature engineering pipeline. An ARIMA(0,0,2) model is applied as a formal forecasting experiment to quantify the index's short-term predictability. Key findings include a total nominal return of 666.9%, a negative Sharpe ratio of -0.45 relative to a 27% Egyptian T-bill risk-free rate, a maximum drawdown of -52.85%, confirmation of volatility clustering, a statistically significant Ramadan effect, and a directional forecasting accuracy of 59.56% — consistent with near weak-form market efficiency. All data, code, and figures are publicly available at: https://github.com/markraouf10/EGX30-Analysis*

Index Terms—*EGX30, Egyptian Exchange, Exploratory Data Analysis, Time Series Analysis, ARIMA, Market Efficiency, Ramadan Effect, Volatility Clustering, Risk Metrics.*

I. INTRODUCTION

The Egyptian Exchange (EGX) is the primary regulated securities market in Egypt, operating under the oversight of the Financial Regulatory Authority (FRA). The EGX30 index, comprising the 30 most actively traded and liquid companies on the exchange, serves as the principal benchmark for Egyptian equity market performance. The index is weighted by free-float market capitalization and reviewed quarterly, with constituents drawn from diverse sectors including banking, real estate, telecommunications, and industrials.

The period from 2014 to 2026 represents one of the most turbulent and economically significant eras in modern Egyptian financial history. Four major macroeconomic events — the November 2016 EGP currency float, the March 2020 COVID-19 market shock, the 2022 currency devaluation, and the 2024 currency crisis — provide rich natural experiments for examining how the EGX30 responds to systemic stress. Additionally, Egypt's unique market calendar (Sunday–Thursday trading week) and cultural calendar (Ramadan) present opportunities to investigate anomalies rarely studied in Western financial literature.

This study makes the following contributions: (1) a comprehensive EDA of EGX30 covering 3,000 trading days with particular attention to fat-tailed return distributions, volatility clustering, and risk-adjusted performance against local risk-free rates; (2) a quantified analysis of the Ramadan effect on Egyptian equity returns and volatility, contributing to the sparse MENA-specific market microstructure literature; (3) an ARIMA forecasting experiment that formally tests whether EGX30 log returns contain statistically exploitable autoregressive structure; and (4) a fully reproducible analysis pipeline with all data, code, and outputs publicly available.

II. RELATED WORK

The efficient market hypothesis (EMH), introduced by Fama [1], posits that asset prices fully reflect available information. The weak form of the EMH implies that historical price data cannot be exploited to generate abnormal returns. Empirical tests of weak-form efficiency in emerging markets have yielded mixed results, with several studies documenting predictable patterns that challenge the hypothesis [2].

Calendar anomalies represent a well-documented class of apparent inefficiencies. The Monday effect — the tendency for the first trading day of the week to underperform subsequent days — has been documented across multiple markets [3]. In Islamic markets where the trading week begins on Sunday, an analogous Sunday effect has been theorized, though empirical evidence remains sparse for the EGX specifically.

The Ramadan effect in Muslim-majority financial markets has been studied by Bialkowski et al. [4], who found evidence of higher returns and lower volatility during Ramadan across several Islamic markets. However, market-specific findings vary considerably, and Egypt's unique combination of high interest rates, currency volatility, and market structure may produce different dynamics than those observed in Gulf Cooperation Council (GCC) markets.

ARIMA models, first systematized by Box and Jenkins [5], remain standard benchmarks for financial time-series forecasting. While machine learning approaches have gained traction, ARIMA provides a transparent, statistically grounded baseline against which more complex models are typically evaluated. For near-efficient markets, ARIMA is expected to produce modest predictive accuracy above random chance, with its structure directly reflecting the degree of exploitable autocorrelation in returns.

III. DATASET AND METHODOLOGY

A. Data Source and Collection

Daily OHLCV (Open, High, Low, Close, Volume) data for the EGX30 index was sourced from Investing.com, covering the period from January 5, 2014 to May 5, 2026 — a total of 3,000 trading days. The Egyptian Exchange operates on a Sunday–Thursday schedule, with approximately 252 trading days per year. Nine missing volume observations were identified and imputed using forward-fill, a standard approach for small gaps in financial time series where the most recent known value serves as a reasonable estimate.

Raw data preprocessing involved: (1) date parsing from MM/DD/YYYY string format to datetime objects; (2) removal of comma-thousands separators from price columns; (3) conversion of abbreviated volume strings (e.g., "496.94M") to raw numeric values; and (4) computation of simple returns and log returns. Log returns were selected as the primary return metric for their desirable statistical properties — time-additivity, approximate symmetry around zero, and superior fit to standard statistical models compared to simple returns.

B. Stationarity Testing

The Augmented Dickey-Fuller (ADF) test was applied to both the closing price series and the log return series. The ADF test evaluates the null hypothesis that a unit root is present (i.e., the series is non-stationary). Results confirmed that closing prices are non-stationary (ADF statistic = 1.41, $p = 1.000$), while log returns are strongly stationary (ADF statistic = −28.38, $p \approx 0.000$), satisfying the prerequisite for ARIMA modeling on the return series.

C. Analytical Framework

The analytical pipeline proceeds through five phases: (1) exploratory data analysis encompassing descriptive statistics, distributional analysis, volatility estimation, drawdown computation, volume analysis, and autocorrelation; (2) calendar anomaly testing for day-of-week and Ramadan effects; (3) risk metric computation including Sharpe ratio, Value at Risk (VaR), and Conditional VaR (CVaR); (4) technical indicator computation and feature engineering; and (5) ARIMA model selection, estimation, and evaluation on a chronological holdout set.

For the Ramadan analysis, Islamic calendar dates for each Ramadan period from 2014 to 2026 were manually specified to ensure accuracy. For the day-of-week analysis, Welch's independent samples t-test was applied to test whether return means differ significantly between trading days. For ARIMA order selection, ACF and PACF plots were examined alongside AIC criteria.

IV. EXPLORATORY DATA ANALYSIS

A. Price History and Major Events

The EGX30 index grew from a level of approximately 6,800 in January 2014 to approximately 52,500 by May 2026, representing a total nominal return of 666.9% and an annualized return of 17.16%. However, this growth was non-linear and punctuated by four significant disruptions.

The November 2016 EGP float caused an immediate sharp nominal appreciation of the index as Egyptian assets became effectively cheaper in hard currency terms and foreign capital began re-entering the market. The COVID-19 shock in March 2020 produced a rapid decline followed by a near-complete V-shaped recovery within the same year. The 2022 devaluation — which occurred in multiple tranches — produced the most severe and sustained drawdown in the dataset, partially driven by the global commodity price shock following the Russian invasion of Ukraine. The 2024 currency crisis produced a sharp decline followed by faster recovery, suggesting markets had developed greater institutional familiarity with the devaluation pattern.

B. Return Distribution Analysis

Descriptive statistics of daily log returns reveal a mean of 0.068% (annualized: 17.16%), a standard deviation of 1.369% (annualized: 21.71%), a worst single-day return of −9.81% (COVID-19 crash, March 2020), and a best single-day return of +6.49%. The distribution exhibits negative skewness (skewness = −0.31), indicating that extreme negative returns are slightly more probable than extreme positive returns of equivalent magnitude.

The excess kurtosis of 5.82 confirms the presence of fat tails (leptokurtosis), indicating that extreme daily moves occur substantially more frequently than a normal distribution would predict. The Jarque-Bera normality test yields a statistic with p-value ≈ 0, conclusively rejecting normality. Q-Q plots confirm systematic deviations at both tails, with the characteristic S-shaped signature of leptokurtic distributions. These findings have direct implications for risk management: standard models assuming Gaussian returns will systematically underestimate the probability of large losses.

C. Volatility Analysis

Rolling 30-day and 90-day annualized volatility were computed as $\sigma\_roll \times \sqrt{252}$, where σ_roll is the rolling standard deviation of log returns. The overall annualized volatility of 21.71% is comparable to other emerging market equity indices.

Distinct volatility regimes are observable in the rolling volatility charts. The COVID-19 shock in March 2020 produced the highest single volatility spike, with 30-day annualized volatility reaching approximately 55%. The 2022 devaluation period sustained elevated volatility (approximately 45% annualized) for an extended period. Inter-crisis periods show substantially lower volatility (15–20% annualized). This clustering behavior is formally tested through autocorrelation analysis of squared returns (see Section IV-F).

D. Drawdown Analysis

The drawdown series, defined as $(P\_t - \max(P\_\tau : \tau \leq t)) / \max(P\_\tau : \tau \leq t)$, quantifies the percentage decline from the running historical peak at each point in time. The maximum drawdown of −52.85% was recorded on July 5, 2022, representing the deepest peak-to-trough loss in the 12-year observation window. An investor who purchased the index at its 2022 peak would have experienced a loss of more than half their capital at the trough — a risk seldom captured by standard volatility metrics alone.

The 2020 COVID drawdown, while severe, recovered within approximately eight months. The 2022 drawdown demonstrated a significantly more prolonged recovery pattern, reflecting the compound structural pressures of devaluation, inflation, and external debt dynamics.

E. Volume Analysis

Daily trading volume exhibited a mean of approximately 182 million EGP and a standard deviation of approximately 129 million EGP. One significant outlier was identified on December 14, 2025, with a single-day volume of 2.8 billion EGP — approximately 15 times the historical average — likely reflecting a major index rebalancing or institutional portfolio event. This observation was retained in the dataset but capped at the 99th percentile for visualization purposes, with explicit annotation.

A Pearson correlation of 0.155 was found between daily trading volume and the absolute value of daily log returns, indicating a weak positive relationship: higher-volume days tend to exhibit larger price movements. This relationship, while statistically significant given the sample size, is insufficiently strong for operational forecasting applications.

F. Autocorrelation Analysis

ACF and PACF analyses were conducted on both log returns and squared log returns. Log returns show negligible autocorrelation at all lags up to 40 — no spike exceeds the 95% confidence bounds after lag 1 — consistent with the near absence of linear predictability in returns. This finding supports the hypothesis that the EGX30 operates near the weak form of market efficiency with respect to past price information.

In contrast, squared log returns exhibit strongly significant autocorrelation at multiple lags, with ACF values decaying slowly. This is the canonical signature of ARCH/GARCH-type volatility clustering: while return direction is unpredictable, return magnitude is persistent. Large-magnitude trading days cluster temporally, particularly around the major economic events identified in Section IV-A.

The Ljung-Box test for autocorrelation in squared returns yields highly significant p-values ($p < 0.001$) at lags 10, 20, and 40, formally confirming volatility clustering. This dichotomy — efficient in returns but clustered in volatility — is a common empirical signature of semi-mature emerging markets.

G. Calendar Effects

Day-of-week analysis was conducted by grouping log returns by the day of the week and computing mean returns. As Egypt's trading week runs Sunday through Thursday, Sunday occupies the position analogous to Monday in Western markets. Results show that Sunday exhibits the lowest mean daily return (−0.095%), while Thursday exhibits the highest (+0.231%). The remaining days show intermediate returns: Monday (+0.019%), Tuesday (+0.129%), and Wednesday (+0.083%).

A Welch's t-test comparing Sunday and Thursday returns yields $t = -3.888$ ($p = 0.0001$), indicating a statistically significant difference. However, a one-sample t-test comparing Sunday returns against zero ($\mu = 0$) yields $t = -1.495$ ($p = 0.135$), confirming that Sunday returns are not significantly negative on their own — only significantly lower than Thursday. This distinction is critical: the finding reflects a relative day-of-week pattern rather than an absolute negative-Sunday phenomenon.

Monthly analysis reveals January as the strongest-performing month on average, with April, May, August, and November also producing positive mean returns. The weakest months include March and September, though these patterns do not reach conventional significance thresholds in multiple-comparison-adjusted tests.

H. Ramadan Effect

Ramadan periods were defined using Islamic lunar calendar dates for each year from 2014 to 2026, encompassing a total of 259 trading days. Non-Ramadan periods account for the remaining 2,741 trading days.

During Ramadan, the mean daily log return is −0.054% compared to +0.080% outside Ramadan. The annualized volatility during Ramadan is 24.3% compared to 21.4% outside — representing both lower returns and higher risk during the holy month. This result contrasts with findings from some Gulf markets where Ramadan is associated with positive sentiment effects, suggesting that Egypt's unique combination of macroeconomic pressures and market structure may override any optimism premium during this period.

The co-occurrence of multiple Ramadan periods with major macroeconomic stress events (e.g., 2016 EGP float, 2020 COVID period) may partially confound the Ramadan effect. Disentangling calendar effects from macroeconomic regime effects would require a longer observation window or conditional analysis beyond the scope of this study.

V. RISK METRICS

The Sharpe ratio, defined as $(R_p - R_f) / \sigma_p$, where R_p is the annualized portfolio return, R_f is the risk-free rate, and σ_p is the annualized volatility, was computed using the Egyptian Central Bank's T-bill rate of approximately 27% per annum as the risk-free benchmark — reflective of the prevailing monetary policy environment throughout much of the analysis period.

The resulting Sharpe ratio of −0.45 indicates that the EGX30 failed to compensate investors for the volatility risk taken, relative to the risk-free alternative available in the local economy. The annualized return of 17.16% fell short of the 27% risk-free rate, making the equity risk premium effectively negative for this period. This result should be interpreted in context: extremely high risk-free rates, driven by monetary tightening to combat inflation and defend the currency, created an unusually demanding benchmark. As monetary policy normalizes, the risk-return dynamic is expected to improve.

Value at Risk at the 95% confidence level (VaR_{95}) is −2.01%, indicating that on 95% of trading days, the index did not lose more than 2.01%. At the 99% confidence level, VaR_{99} = −4.12%. The Conditional VaR ($CVaR_{95}$), also known as Expected Shortfall, is −3.27% — representing the average loss on the worst 5% of trading days. CVaR provides a more complete picture of tail risk than VaR alone, as it describes the expected severity of losses beyond the threshold rather than merely the threshold itself.

VI. FEATURE ENGINEERING

A total of 33 features were constructed for the ARIMA baseline and potential future modeling. Technical indicators included Simple Moving Averages (SMA) and Exponential Moving Averages (EMA) at windows of 7, 20, 50, and 200 days, providing trend signals at multiple time horizons. The Relative Strength Index (RSI, 14-day) was computed as a momentum oscillator, the Moving Average Convergence Divergence (MACD) as a trend-momentum indicator, and Bollinger Bands (20-day, ±2σ) as volatility-normalized price envelopes.

The Average True Range (ATR, 14-day) was included as a pure volatility measure incorporating overnight price gaps. Lagged return features were constructed at lags of 1, 2, 3, 5, and 10 days to capture potential short-term autocorrelation. Rolling volatility at 10-day and 30-day windows was included as a regime indicator. Volume was normalized by its 20-day moving average to produce a volume ratio feature robust to secular trends in market size.

A Pearson correlation heatmap of the primary analytical features confirms minimal redundancy: log returns show near-zero correlation with most technical indicators (as expected from the ACF analysis), while volatility and Bollinger Band width measures exhibit moderate positive correlations consistent with their shared construction from return variance.

VII. ARIMA FORECASTING

A. Model Selection

ARIMA model order selection followed a two-step procedure. First, the ACF and PACF of log returns were examined: rapid decay to zero at all lags confirmed that low-order AR and MA terms, if any, would be appropriate. Second, an initial ARIMA(2,0,2) model was estimated; statistical significance tests on individual coefficients revealed that both AR terms were statistically insignificant (AR(1): $p = 0.69$, AR(2): $p = 0.16$), while both MA terms approached significance (MA(1): $p = 0.000$, MA(2): $p = 0.013$). The model was reduced to ARIMA(0,0,2) — a pure MA(2) specification — consistent with the ACF/PACF evidence showing no meaningful autoregressive structure.

The differencing order $d = 0$ was confirmed by the ADF test result: log returns are stationary without differencing. The Akaike Information Criterion (AIC) favored the reduced ARIMA(0,0,2) over the initial ARIMA(2,0,2), validating the parsimony of the specification.

B. Estimation and Evaluation

The ARIMA(0,0,2) model was estimated on a training set comprising the first 85% of the log return series (approximately 2,543 observations, January 2014 – November 2024), with the remaining 15% (approximately 448 observations, November 2024 – May 2026) reserved as a held-out test set. Strict chronological ordering was maintained throughout — no shuffling was applied, as shuffling would introduce data leakage by exposing the model to future information during training.

On the test set, the model achieves: Root Mean Squared Error (RMSE) = 0.01118, Mean Absolute Error (MAE) = 0.00830, and directional accuracy = 59.56% — representing the proportion of test-period trading days where the model correctly predicted the sign of the next-day return.

The directional accuracy of 59.56% exceeds the 50% random baseline by approximately 9.56 percentage points, confirming that the model captures modest but real short-term predictive structure in returns. However, the MA(2) structure — the only significant component — implies that this structure is limited to the effects of past two-day forecast errors rather than meaningful autoregressive patterns, consistent with market near-efficiency.

C. Residual Diagnostics

Ljung-Box tests on model residuals yielded p-values of 0.638 (lag 10) and 0.246 (lag 20), both well above the 0.05 threshold. Failure to reject the null hypothesis of no residual autocorrelation confirms that the model adequately captured the exploitable linear structure in the series — residuals constitute white noise. The residual distribution exhibits the same fat-tailed characteristics as raw returns (kurtosis > 3), consistent with the Jarque-Bera rejection of normality observed in Section IV-B. This suggests that while the ARIMA model is correctly specified for the conditional mean, a GARCH-class extension would be needed to model the conditional variance.

VIII. RESULTS AND DISCUSSION

Table I summarizes the key quantitative findings of this study.

Metric	Value
Total Return (12 yrs)	666.9%
Annualized Return	17.16%
Annualized Volatility	21.71%
Sharpe Ratio (RF=27%)	−0.45
Max Drawdown	−52.85% (Jul 5, 2022)
VaR 95% (daily)	−2.01%
CVaR 95% (daily)	−3.27%
Skewness	−0.31
Excess Kurtosis	5.82
ARIMA Model	ARIMA(0,0,2)

Directional Accuracy	59.56%
Ljung-Box p (lag 10)	0.638
Ramadan Avg Return	−0.054%
Non-Ramadan Avg Return	+0.080%
Sunday Avg Return	−0.095%
Thursday Avg Return	+0.231%

TABLE I. Summary of Key Quantitative Findings

The central tension in these findings is between nominal and risk-adjusted performance. The 666.9% total return and 17.16% annualized return represent substantial nominal wealth creation; however, relative to the 27% risk-free rate available from Egyptian T-bills throughout much of this period, the equity risk premium was negative. This does not indicate that the EGX30 is fundamentally a poor investment vehicle, but rather that the specific macroeconomic environment — characterized by repeated currency devaluations, high monetary policy rates, and persistent inflation — created an unusually demanding benchmark for equity returns during this window.

The ACF/PACF analysis and ARIMA results together paint a consistent picture: the EGX30 is near weak-form efficient with respect to historical price data. Return autocorrelation is negligible, and the ARIMA model's 59.56% directional accuracy, while statistically above chance, is insufficient to support profitable trading strategies after accounting for transaction costs and slippage. The market is not perfectly efficient — the MA(2) structure indicates some residual predictability — but the exploitable component is too small for practical use.

The Ramadan finding is noteworthy as an original contribution. Unlike some Gulf markets where Ramadan coincides with positive return effects, the EGX30 shows negative average returns and elevated volatility during Ramadan. This may reflect the specific operational realities of the Egyptian market — shorter trading hours reducing liquidity, reduced institutional activity, and the correlation between Ramadan timing and historical macroeconomic stress events — rather than a universal Islamic market effect.

IX. CONCLUSION

This paper presented a comprehensive 12-year historical analysis of the EGX30 index, combining exploratory data analysis, risk metric computation, technical feature engineering, and ARIMA forecasting to characterize the statistical properties and predictability of the Egyptian equity market.

Five principal conclusions emerge from the analysis: (1) The EGX30 delivered strong nominal returns (666.9% total) but negative risk-adjusted returns (Sharpe = −0.45) relative to the Egyptian risk-free rate, reflecting the challenging macroeconomic context of the analysis period. (2) Return distributions exhibit significant non-normality with fat tails (kurtosis = 5.82) and negative skewness, implying that standard Gaussian risk models substantially underestimate actual risk. (3) Volatility clustering is statistically confirmed via ACF of squared returns, with volatility concentrating around major economic events — particularly the 2022 devaluation, which produced the maximum drawdown of −52.85%. (4) Calendar anomalies — including a significant Sunday underperformance relative to Thursday and a Ramadan-associated return and volatility differential — are statistically present but practically unexploitable after transaction costs. (5) The ARIMA(0,0,2) model achieves 59.56% directional accuracy, confirming near but not perfect weak-form efficiency.

Future research directions include GARCH-class modeling of conditional volatility, event study analysis with abnormal return calculations around specific devaluation dates, machine learning approaches to multi-feature return prediction, and cross-market comparison with regional indices such as the Saudi Tadawul and UAE DFM to contextualize EGX30 performance within the broader MENA equity landscape.

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